Aggregation in SQL

Question 1: Count how many cities are there in each country?

SQL Query:

```
USE world;

SELECT co.Name AS CountryName,
       COUNT(ci.ID) AS TotalCities
FROM country co
LEFT JOIN city ci
   ON co.Code = ci.CountryCode
GROUP BY co.Code, co.Name
ORDER BY co.Name;
```

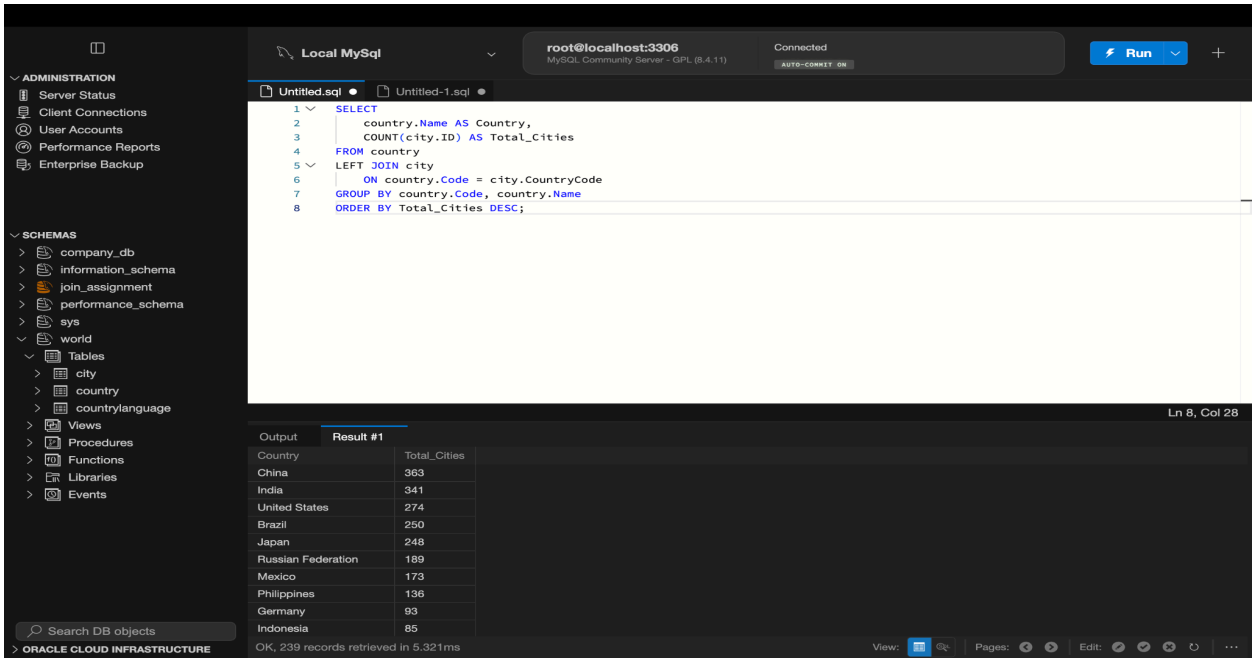
Explanation:

The `COUNT()` function counts the cities in each country. The `LEFT JOIN` includes all countries, even those without any cities in the database. The `GROUP BY` clause groups the results by country.

Result:

The query displays each country name and the total number of cities recorded for that country.

Practical proof



Question 2: Display all continents having more than 30 countries.

SQL Query:

```
SELECT Continent,
      COUNT(*) AS TotalCountries
FROM country
GROUP BY Continent
HAVING COUNT(*) > 30;
```

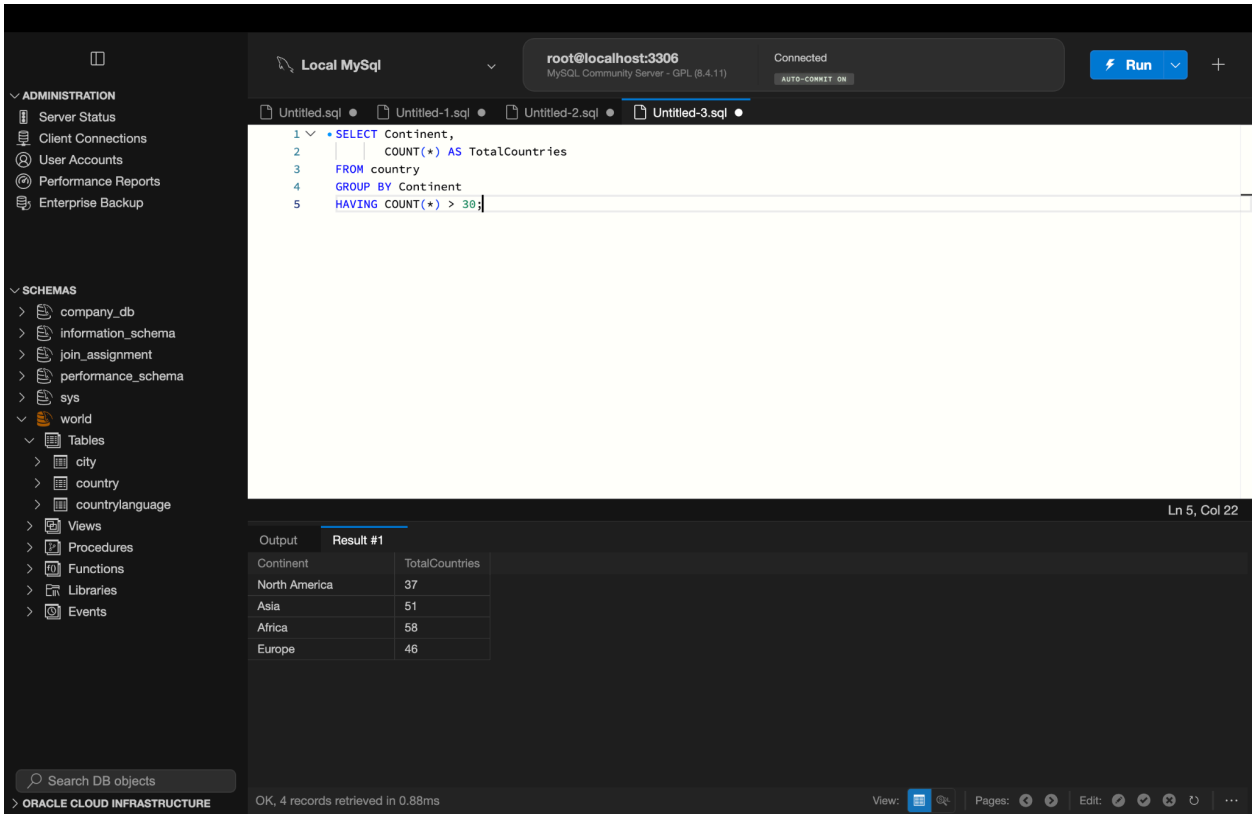
Explanation:

The `GROUP BY` clause groups countries by continent. The `COUNT()` function counts the countries in each continent, and the `HAVING` clause displays only continents with more than 30 countries.

Result:

The query displays the continents that contain more than 30 countries in the World database, along with their country counts.

Practical proof



Question 3: List regions whose total population exceeds 200 million.

SQL Query:

```
SELECT Region,
    SUM(Population) AS TotalPopulation
FROM country
GROUP BY Region
HAVING SUM(Population) > 200000000
ORDER BY TotalPopulation DESC;
```

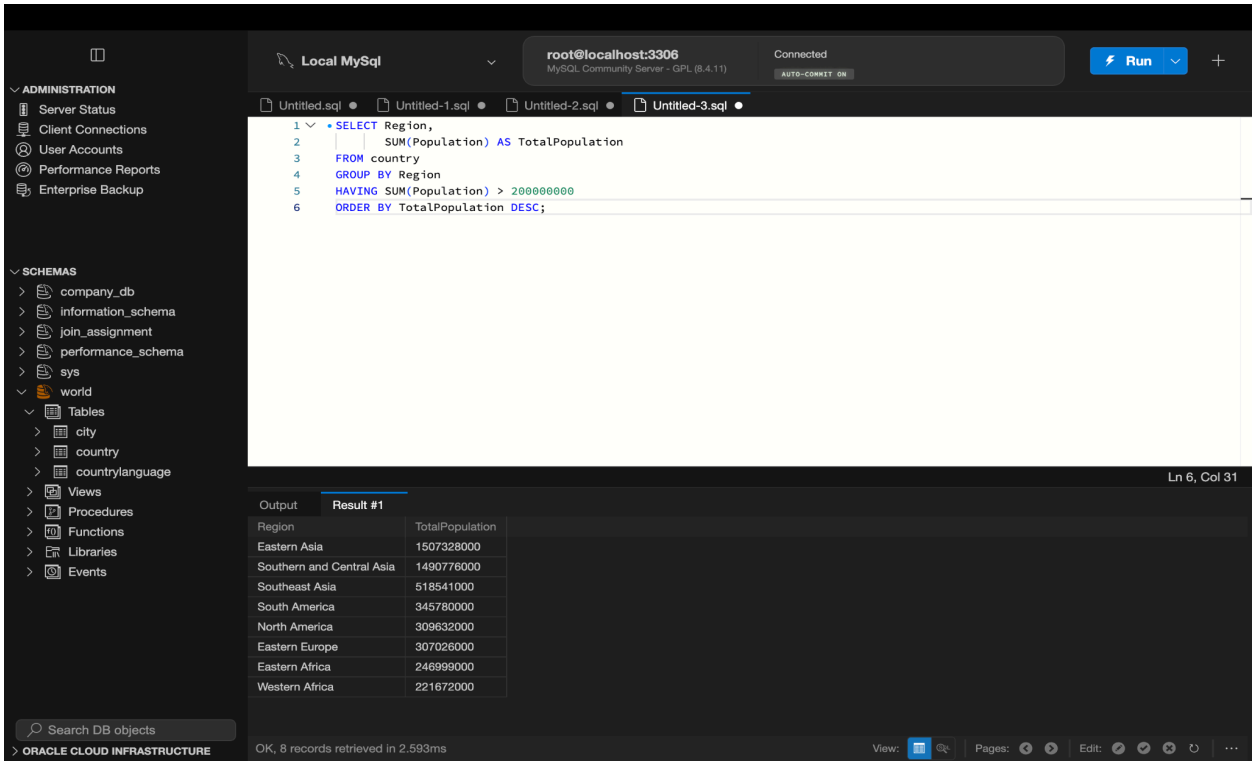
Explanation:

The `SUM()` function calculates the total population of countries within each region. The `GROUP BY` clause groups countries by region, while the `HAVING` clause filters regions whose total population exceeds 200 million.

Result:

The query displays the regions with a combined population greater than 200 million, ordered from highest to lowest population.

Practical proof :



Question 4: Find the top 5 continents by average GNP per country.

SQL Query:

```
SELECT Continent,
      AVG(GNP) AS AverageGNP
FROM country
WHERE GNP IS NOT NULL
GROUP BY Continent
ORDER BY AverageGNP DESC
LIMIT 5;
```

Explanation:

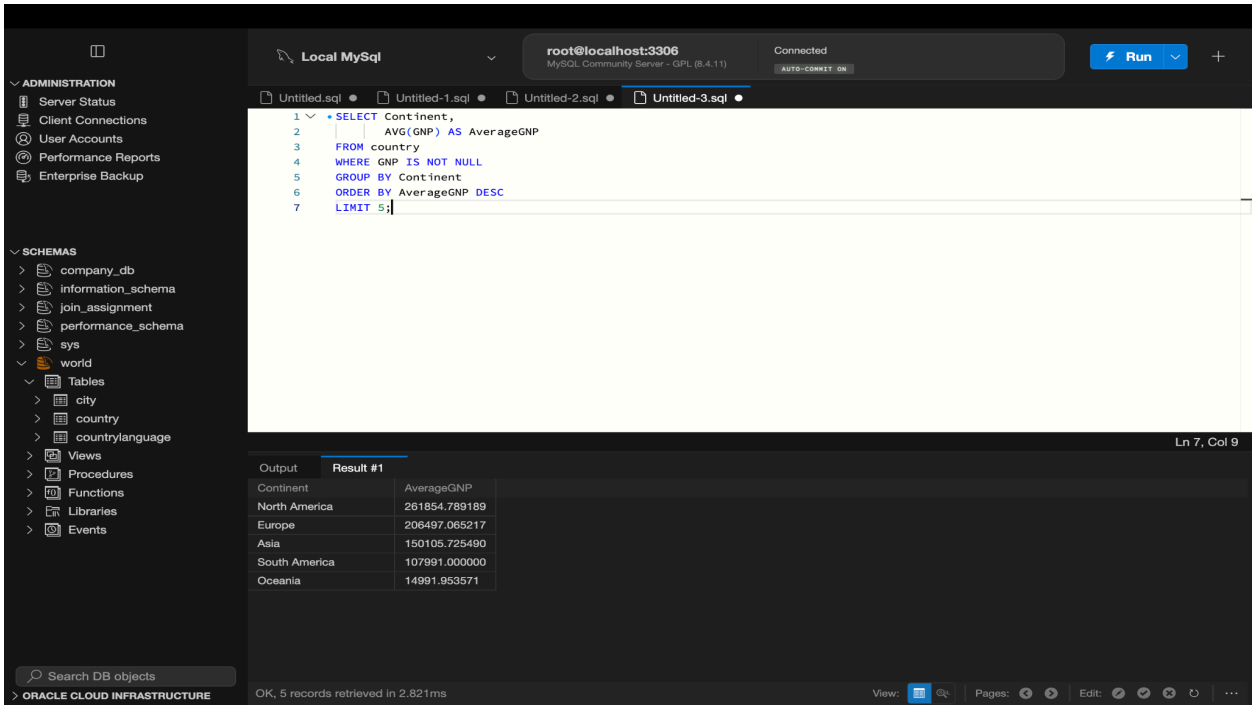
The `AVG()` function calculates the average GNP of countries in each continent. The `WHERE` clause excludes countries with missing GNP values. The `ORDER BY` clause sorts

the results from highest to lowest average GNP, and `LIMIT 5` returns the top five continents.

Result:

The query displays the five continents with the highest average GNP per country.

Practical proof :



Question 5: Find the total number of official languages spoken in each continent.

SQL Query:

```
SELECT c.Continent,
       COUNT(cl.Language) AS TotalOfficialLanguages
FROM country c
JOIN countrylanguage cl
    ON c.Code = cl.CountryCode
WHERE cl.IsOfficial = 'T'
GROUP BY c.Continent
ORDER BY TotalOfficialLanguages DESC;
```

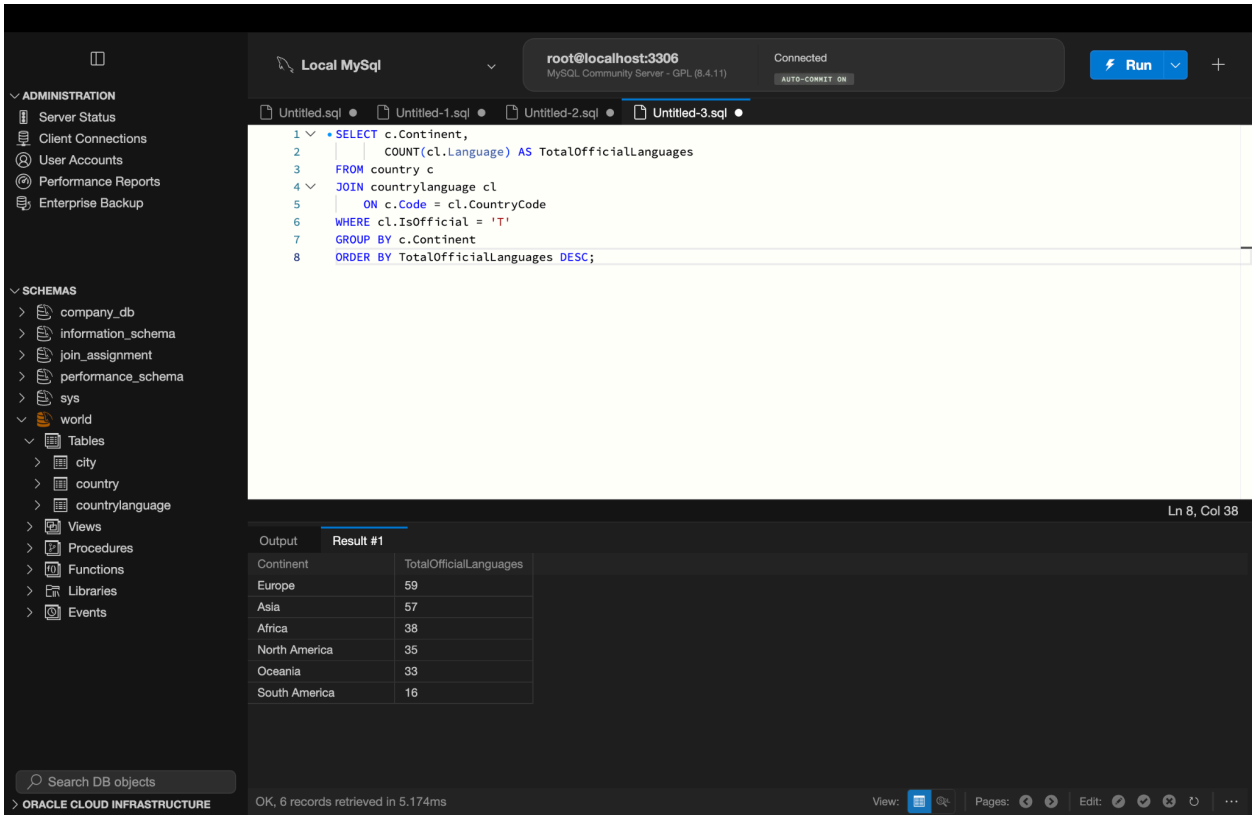
Explanation:

The query joins the `country` and `countrylanguage` tables using the country code. The `WHERE` clause selects only official languages. The `COUNT()` function counts official-language records for each continent, and `GROUP BY` groups the results by continent.

Result:

The query displays each continent and the total number of official-language records associated with its countries.

Practical proof :



Question 6: Find the maximum and minimum GNP for each continent.

SQL Query:

SELECT Continent,

MAX(GNP) AS MaximumGNP,

MIN(GNP) AS MinimumGNP

FROM country

WHERE GNP IS NOT NULL

GROUP BY Continent;

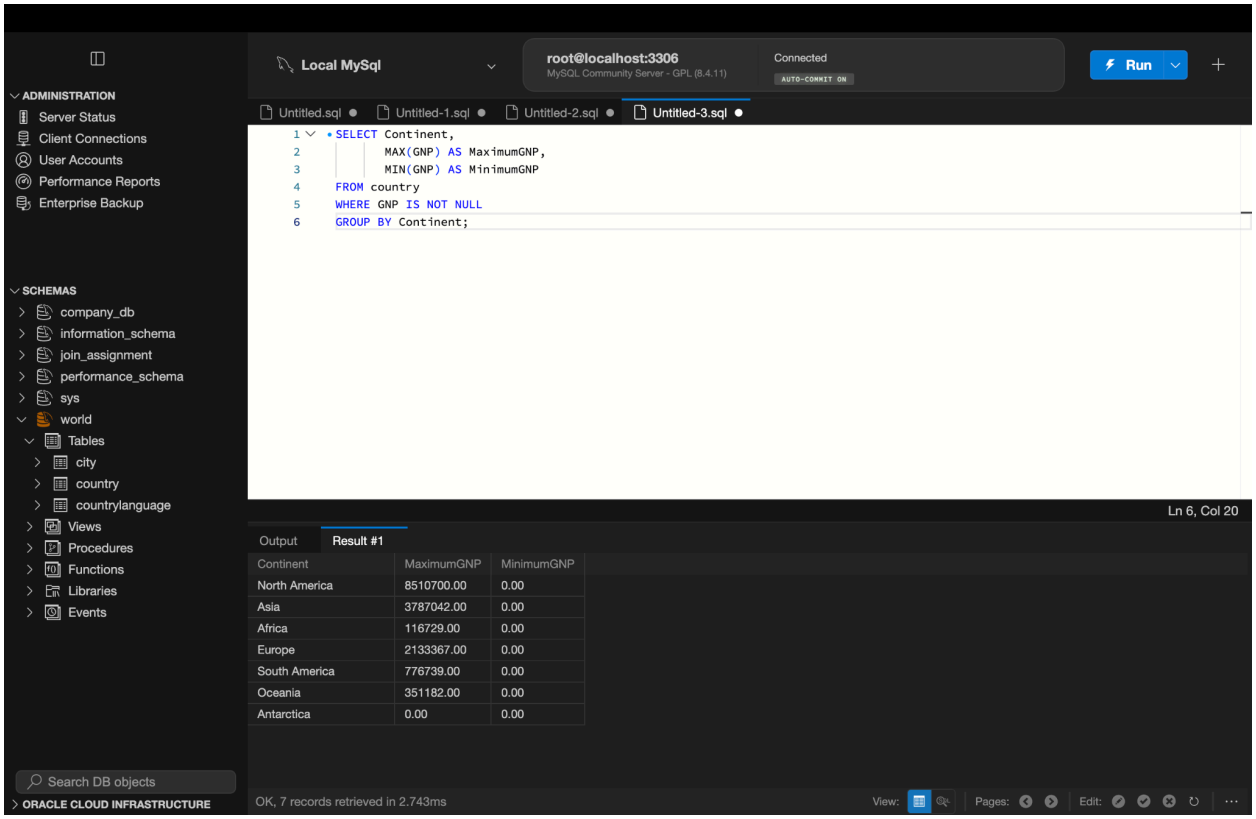
Explanation:

The `MAX()` function identifies the highest GNP, while the `MIN()` function identifies the lowest GNP within each continent. The `GROUP BY` clause groups countries by continent. Countries with missing GNP values are excluded.

Result:

The query displays the maximum and minimum GNP for each continent.

Practical proof :



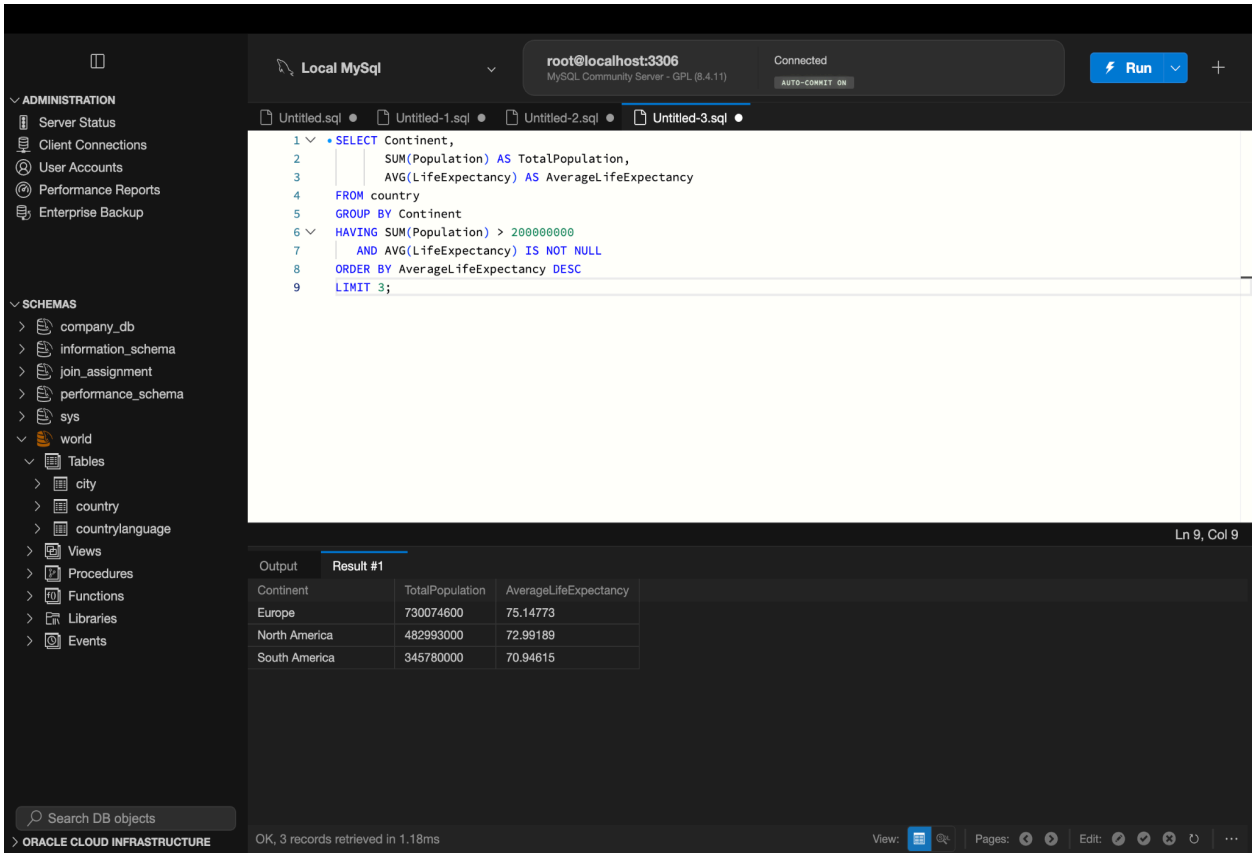
Question 7: Find the country with the highest average city population.

SQL Query:

SELECT co.Name AS CountryName,

AVG(ci.Population) AS AverageCityPopulation

FROM country co



JOIN city ci

ON co.Code = ci.CountryCode

GROUP BY co.Code, co.Name

ORDER BY AverageCityPopulation DESC

LIMIT 1;

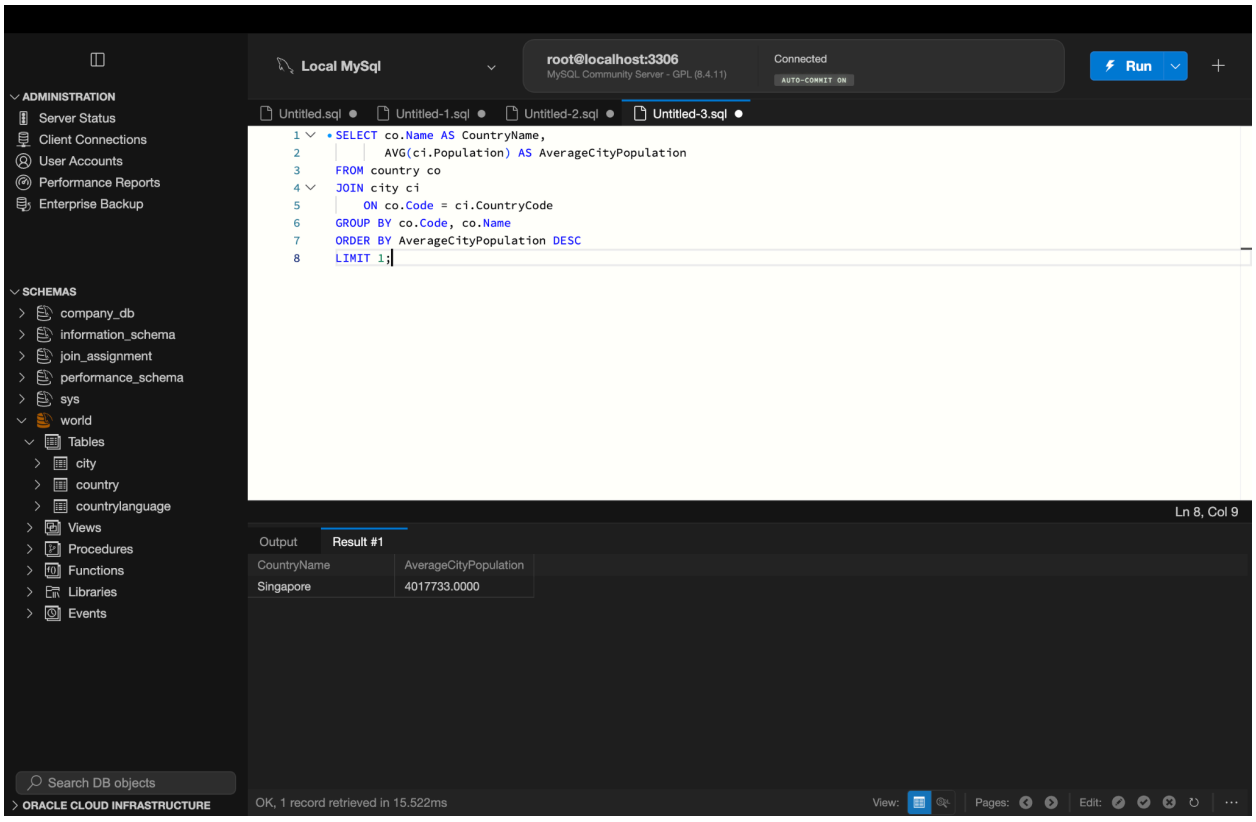
Explanation:

The query joins the `country` and `city` tables using the country code. The `AVG()` function calculates the average population of cities in each country. The `ORDER BY` clause sorts the countries by average city population in descending order, and `LIMIT 1` returns the country with the highest average.

Result:

The query displays the country with the highest average city population and its average city population.

Practical proof :



Question 8: List continents where the average city population is greater than 200,000.

SQL Query:

SELECT co.Continent,

AVG(ci.Population) AS AverageCityPopulation

FROM country co

JOIN city ci

ON co.Code = ci.CountryCode

GROUP BY co.Continent

HAVING AVG(ci.Population) > 200000

ORDER BY AverageCityPopulation DESC;

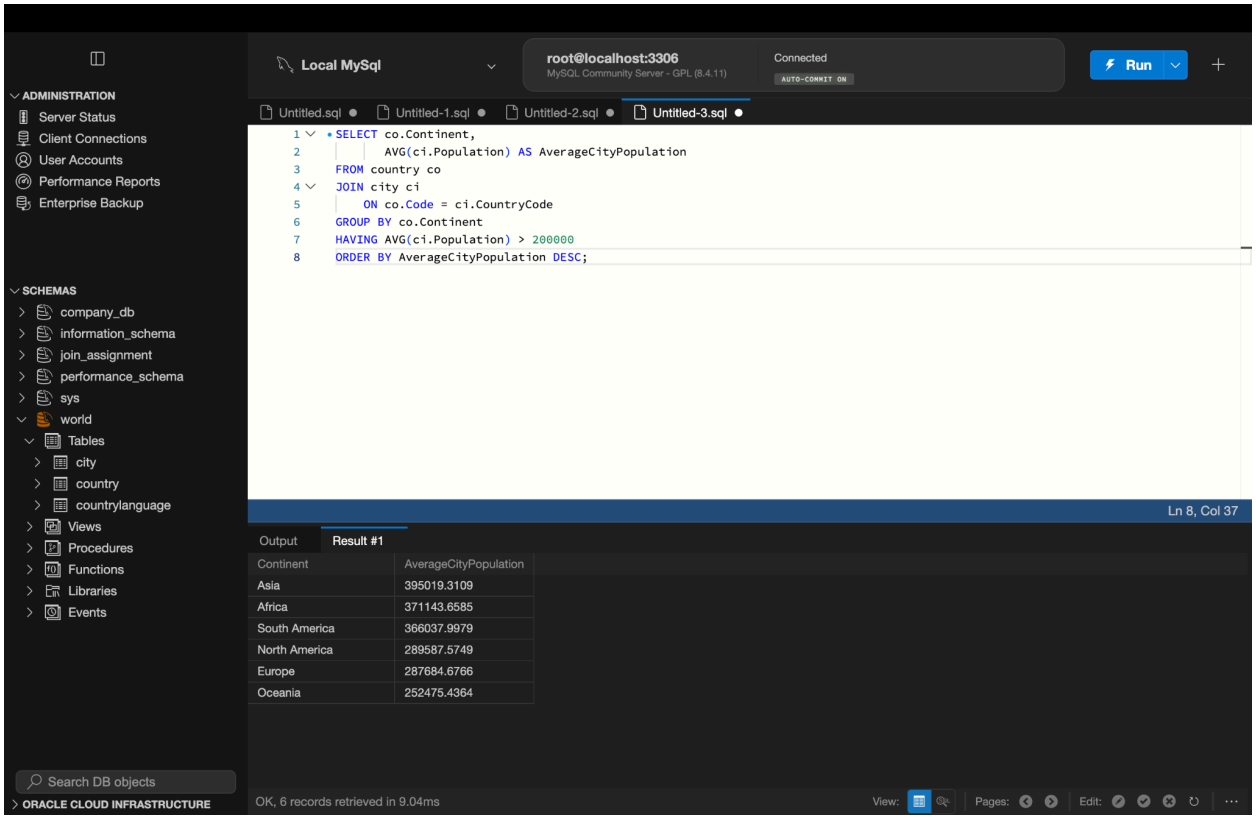
Explanation:

The query joins the `country` and `city` tables. The `AVG()` function calculates the average population of cities within each continent. The `GROUP BY` clause groups the results by continent, and the `HAVING` clause selects only continents with an average city population greater than 200,000.

Result:

The query displays the continents whose average city population exceeds 200,000, ordered from highest to lowest average.

Practical proof :



Question 9: Find the total population and average life expectancy for each continent, ordered by average life expectancy descending.

SQL Query:

```
SELECT Continent,
    SUM(Population) AS TotalPopulation,
    AVG(LifeExpectancy) AS AverageLifeExpectancy
FROM country
GROUP BY Continent
ORDER BY AverageLifeExpectancy DESC;
```

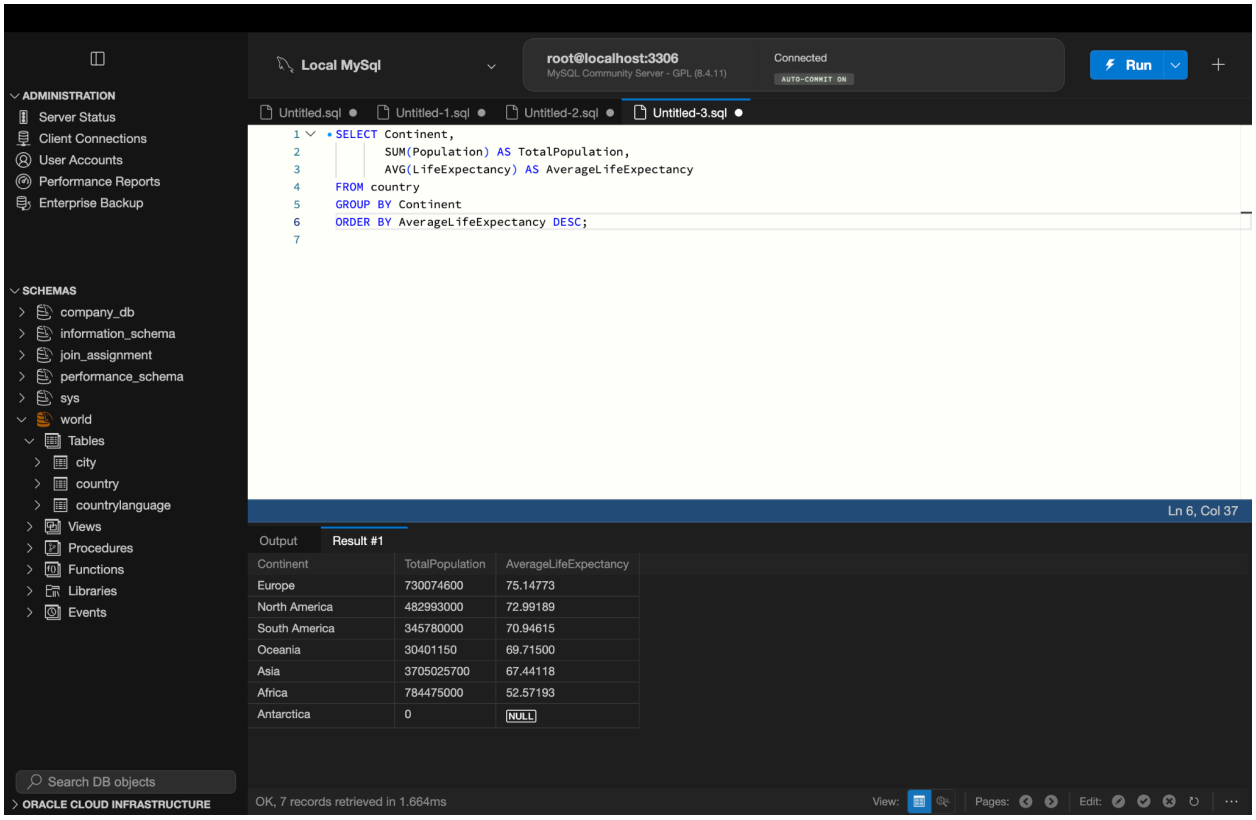
Explanation:

The `SUM()` function calculates the total population of each continent. The `AVG()` function calculates the average life expectancy of countries within each continent, ignoring NULL values. The `GROUP BY` clause groups the results by continent, and the `ORDER BY` clause sorts the results by average life expectancy in descending order.

Result:

The query displays each continent's total population and average life expectancy, ordered from highest to lowest average life expectancy.

Practical proof :



Question 10: Find the top 3 continents with the highest average life expectancy, but only include those where the total population is over 200 million.

SQL Query:

```
SELECT Continent,
    SUM(Population) AS TotalPopulation,
    AVG(LifeExpectancy) AS AverageLifeExpectancy
FROM country
GROUP BY Continent
HAVING SUM(Population) > 200000000
  AND AVG(LifeExpectancy) IS NOT NULL
```

ORDER BY AverageLifeExpectancy DESC

LIMIT 3;

Explanation:

The `SUM()` function calculates the total population of each continent, while the `AVG()` function calculates its average life expectancy. The `HAVING` clause includes only continents with a total population greater than 200 million and a non-NULL average life expectancy. The `ORDER BY` clause sorts the continents by average life expectancy in descending order, and `LIMIT 3` returns the top three.

Result:

The query displays up to three continents with a total population greater than 200 million and the highest average life expectancy.

Practical proof :